

Introduction to Data Analytics

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Introduction

Introduction

Definition

Data Analytics is the process of examining raw data to discover useful information, patterns, and insights.

Explanation

It helps organizations make better decisions using collected data.

Example

Amazon recommends products based on previous searches and purchases.

What is Data?

Definition

Data refers to raw facts, figures, or observations collected for analysis.

Types of Data

Numerical Data

Text Data

Image Data

Video Data

Example

Student marks, mobile numbers, attendance records.

Importance of Data

Why Data is Important?

Helps in decision making

Improves business performance

Identifies customer needs

Predicts future trends

Real-Life Example

Hospitals use patient data to improve treatment.

Introduction to Data Analytics

Definition

Data Analytics converts raw data into meaningful information.

Main Goal

To identify patterns and support decision-making.

Applications

Banking

Healthcare

Education

E-commerce

Why Data Analytics is Important?

Importance

Better business decisions

Cost reduction

Customer satisfaction

Fraud detection

Increased productivity

Example

Banks detect fraud transactions using analytics.

Data Analytics Process

Steps in Data Analytics

Data Collection

Data Cleaning

Data Processing

Data Analysis

Data Visualization

Decision Making

Example

Analyzing sales data to improve profit.

Types of Data Analytics

Four Types

Descriptive Analytics

Diagnostic Analytics

Predictive Analytics

Prescriptive Analytics

Purpose

Each type solves different business problems.

Descriptive Analytics

Definition

Explains what happened in the past.

Example

Monthly sales reports.

Tools

Excel

Power BI

Tableau

Diagnostic Analytics

Definition

Explains why something happened.

Example

Finding reasons for low sales.

Techniques

Drill-down analysis

Data mining

Predictive Analytics

Definition

Predicts future outcomes using historical data.

Example

Weather forecasting.

Techniques

Machine Learning

Regression

Prescriptive Analytics

Definition

Suggests the best action to take.

Example

Google Maps suggesting shortest route.

Benefits

Improves decision making.

Data Analysis Vs Data Analytics

Data Analysis

Studies existing data

Focus on past

Small scope

Reporting

Data Analytics

Complete data process

Focus on future

Large scope

Prediction & decision

Classification of Data Analytics

Based on Industry

Business Analytics

Healthcare Analytics

Financial Analytics

Social Media Analytics

Based on Technique

Statistical Analytics

Machine Learning Analytics

Elements of Data Analytics

Main Elements

Data Collection

Data Storage

Data Cleaning

Data Processing

Visualization

Reporting

Importance

These elements improve data quality.

Data Analyst Vs Data Scientist

Data Analyst

Creates reports

Uses Excel & SQL

Focus on analysis

Data Scientist

Builds AI models

Uses Python & ML

Focus on prediction

Applications of Data Analytics

Applications

Healthcare

Banking

Education

Retail

Transportation

Examples

Fraud detection

Recommendation systems

Disease prediction

Advantages of Data Analytics

Advantages

Faster decisions

Better accuracy

Business growth

Customer understanding

Cost reduction

Challenges in Data Analytics

Challenges

Large data volume

Data privacy issues

Poor data quality

Security risks

High implementation cost

Tools Used in Data Analytics

Popular Tools

Python

R

SQL

Excel

Power BI

Tableau

Why Python?

Easy syntax

Powerful libraries

Large community support

Conclusion

Summary

Data Analytics helps convert raw data into useful insights.

Different analytics types solve different problems.

Widely used in modern industries.

Final Statement

Data Analytics is an essential technology for business growth and decision-making.

References

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Data Science and Big Data Analytics – Wiley Publications

Statistics 101 – David Borman

Machine Learning – Kevin P. Murphy

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